



Niagara II Processor

What is it?

'Niagara' is the code name for the UltraSPARC T1 processor. It is a multi-core, multi-threaded CPU. The next generation, UltraSPARC T2 processor is sometimes called Niagara II.

Who is behind it?

Sun Microsystems. Niagara was their third multi-core processor, but the first multi-core, multi-threaded processor. It came in three versions: four, six or eight CPU core.

Why is it important?

Each of the cores can process eight threads simultaneously. So with eight cores, that's 64 threads that can be processed concurrently!

Where will it be used?

Servers. Sun is targeting the \$100 billion a year server market, in which Sun and IBM are the largest players. The beauty of multi-threading is that you get a lot more throughput with a much lower clock speed. Also, Sun is advertising them as eco-friendly processors.

When is it releasing?

Reports indicate that it will be released in the second half of 2007. The processor will be manufactured using a 65 nanometer process.

How much better will Niagara II be?

Niagara II will improve mathematical abilities by adding more floating-point calculation units. Currently, a single FPC unit is shared by all eight cores. Networking will also be beefed up, thanks to the built in 10 Gbps Ethernet. If that's not enough, Sun plans to ship the UltraSPARC T2 (Niagara II) as dual-processors, so that's 16 cores, and 128 simultaneous threads being computed!

flatlining community", and that laptops were generating a little more excitement.

The basis for these claims is the adoption of technology in developing countries, who are bypassing the efforts of setting up elaborate wired infrastructures in favour of going wireless, leading to a world where everyone owns a smartphone. A classic example, they say, is India, where while PC adoption increases by 5 million a year, mobile phone adoption increases by that much a month. The smartphone will set people free from laptops, says head of propositions John Forsyth—poor battery performance will inevitably give way. With mobile phone screens getting better with every new generation, productivity and readability on the mobile will supposedly reach new heights, leaving people wondering why they need a PC at all.

This is, of course, the millionth time that the death of the PC has been touted—PDAs have been "threatening" the PC since the mid-nineties, as have Java-based thin-client-like terminals. While we don't see anything replacing the PC per se, one must accept that shifting your work to your handheld device is a lot easier today than it's ever been. With the huge number of Internet-based applications, all you need to get your work done is a connection. So it's quite possible that the Symbian execs aren't too far from the truth—at least about possible growth of the smartphone.

In other unrelated news, research and analysis firm Gartner has published a report saying that PDAs and smartphones aren't growing at the rate that they were expected to, forcing them to revise their year-end forecasts. The report states that

smartphone adoption seems to have more to do with flaunt value than functionality, and that phones with less features, but better aesthetics will do better in the market.

So will the supposedly flagging smartphone really kick the desktop into oblivion? You decide!

A SECOND REUTER'S

Newsdesk In A Parallel Universe

Media giant Reuter's has announced that they have opened a news bureau within Second Life (www.secondlife.com), the massive simulation game that has become something of an alternative universe on the Internet. The news bureau will bring news from the virtual world to readers in

world, complete with cities, localities, and characters, where users can build characters and homes, and earn and trade in the virtual currency—the Linden Dollar.

Think of it as a large-scale version of SimCity and The Sims rolled into one. There's even an online marketplace where users can turn their Linden Dollars into real US dollars. In the 24 hours preceding our writing this, \$447,446 (that's in real US dollars) had been spent by users of the game.

With this alarming popularity, it's no surprise that companies have started to invade this space as well.

Sun recently set up its own Second Life space to promote its Project Darkstar—which will simplify the development of multiplayer and massively multiplayer games. Media giant CNET too has their own Second Life presence.



the real one and vice versa, and can be found at secondlife.reuters.com.

Second Life, created by Linden Labs, has become an online phenomenon in a way that few things ever have. As of this writing, it has seen over a million registrations, with over four hundred thousand active within the last two months. The concept isn't really new—an online

If you thought MMORPGs were even remotely addictive, the mind boggles at the potential of a simulation at this scale—we haven't encountered any reports of people dropping dead at their desks while Second Life-ing, so that's a good sign, for now!

An aside: would you like a Second Digit? Write in and let us know! ■